
QS 501P: Experiments in Quantum Optics

Course Code	QS 501P
Course Name	Experiments in Quantum Optics
L-T-P-C	0-0-5-3
Prerequisites	First course on Quantum Mechanics
Intended for	B.Tech, M.Tech, M.Sc, Ph.D.
Approval	54th BoA (May 2024)

Course Contents

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Laboratory Modules

1. Fourier Transform: To demonstrate the Fourier Transform of sample images
2. Optical Tweezers: Demonstrate microscale particle manipulation dynamics through laser induced forces using Portable Optical Tweezers.
3. Optical Vortices: Demonstrate the Optical Vortices corresponding to various topological charges.
4. Quantum Eraser: To demonstrate the sensitivity of path information on the measurement process
5. Polarized 3D Cinema: Create a 3D Projection using circularly polarized light based on Real 3D Cinema Technology
6. Bomb Tester: Experiment on Interaction-free Quantum measurements: Testing the Bombs in Michelson Interferometer
7. Quantum Cryptography: Demonstration of BB84 Quantum Cryptography Protocol and detection of eavesdropper on Encrypted data

Text Books

1. Christopher Garry, **Introductory Quantum Optics**, Cambridge University Press, 2012.
2. Mark Fox, **Quantum Optics: An Introduction, Oxford master series in Physics**, 2006.
3. Laboratory manual for experiments

References

1. NA